

USING SATELLITE DATA TO IMPROVE CLIMATE MODELS:
CORRECTING SEA LEVEL AND WAVE HEIGHT PROJECTIONS
ISRO RESEARCH GRANT PROJECT — PROGRESS UPDATE (MAR 2025–JAN 2026)

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AGENDA

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INTRODUCTION

WHY THIS MATTERS

- To keep coastal communities safe, we need accurate predictions of sea levels and waves.
- Current climate models (CMIP6) give long-term forecasts but often have errors.
- Satellite data provides a reliable "ground truth" to measure and fix these errors.

WHAT WE MEASURE

- Significant Wave Height (SWH) and Sea Level Anomaly (SLA).
- Focus: Indian Ocean; using overlapping data periods to train our models.

RESEARCH PROBLEM AND OBJECTIVES

THE PROBLEM

- Climate models struggle with local details, causing errors that vary by location and season.
- If uncorrected, these errors make it hard to predict future risks like floods or storms.

OUR OBJECTIVES

- A Quantifying the uncertainties in the projections of significant wave height and sea level anomalies available from CMIP6 data during the observed period (2015–2023).
- B Developing an AI/ML-based model to correct for the uncertainties in the CMIP6 dataset.
- C Generating the probabilistic long-term forecast with high reliability for the significant wave height and sea level over the Indian Ocean using the modified projections of CMIP6 datasets.

LITERATURE REVIEW

WHAT IS ESTABLISHED

- Traditional statistical fixes often fail when errors are complex or changing over time.
- Machine Learning is better at finding and fixing these complex error patterns.
- Satellites give us the best global coverage to check these ocean models.

WHY OUR APPROACH WORKS

We use satellite data to guide a Machine Learning model, creating a scalable way to fix climate predictions.

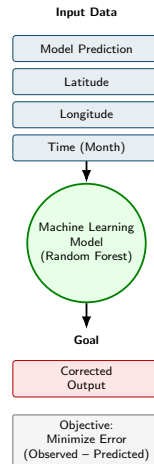
METHODOLOGY

DATA AND PREPROCESSING

- Observations: Combined satellite data (SWH, SLA).
- Models: CMIP6 historical data + Future scenarios.
- Steps: Monthly averages, quality checks, and matching model grids to satellite data.

CORRECTION AND EVALUATION

- Inputs: Model prediction + Location (Lat/Lon) + Time.
- Training Split: 2015–2018 for training; 2019–2020 for testing.



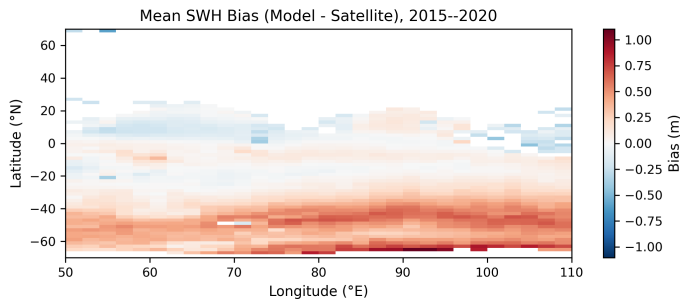
WORK DONE

- Gathered and organized data from multiple satellites for Sea Level and Wave Height.
- Processed CMIP6 climate model data to match the satellite format.
- Built and tested the first version of our Machine Learning correction tool.
- Checked how well it works on new data it hasn't seen before (2019–2020).

SPATIAL BIAS ANALYSIS

BIAS DISTRIBUTION

- Map shows where the model overestimates (red) or underestimates (blue) wave heights.
- High Bias Regions:** Arabian Sea and Southern Ocean show strong overestimation.
- Implication:** A single correction factor isn't enough; we need location-specific corrections.



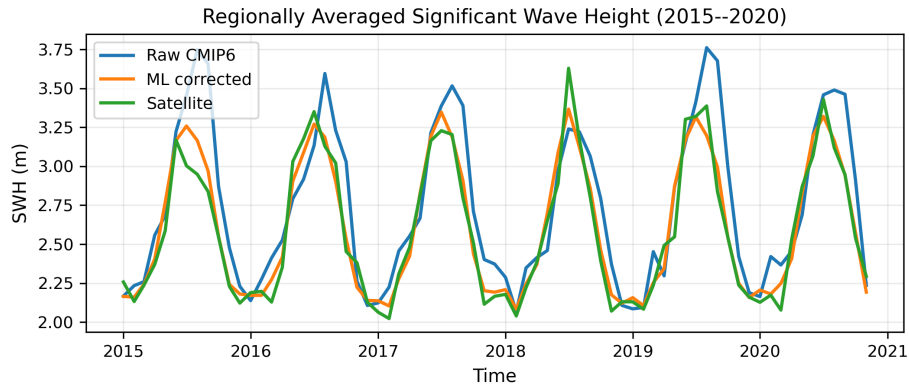
DETAILED PERFORMANCE METRICS

MODEL PERFORMANCE ON INDEPENDENT TEST SET (2019–2020)

Metric	Raw CMIP6	Corrected	Improvement
RMSE (m)	0.610	0.408	33.2%
Bias (m)	+0.179	−0.013	92.7%
R^2 Score	0.692	0.880	27.2%
MAE (m)	0.456	0.293	35.7%

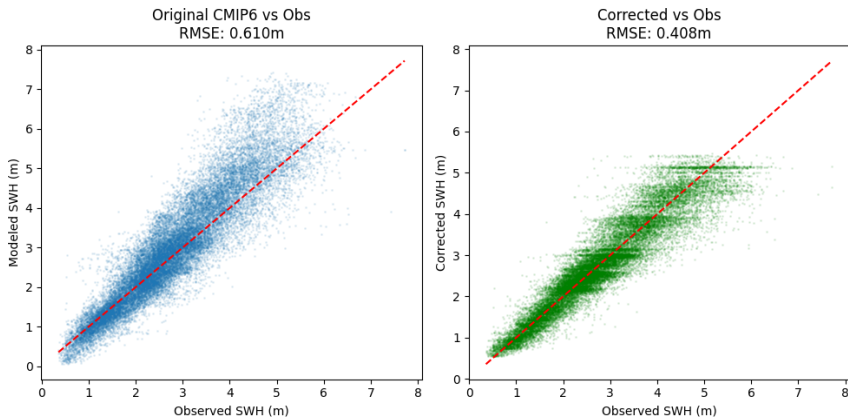
- **RMSE reduced by 33%**, indicating significantly better accuracy.
- **Bias eliminated** (near zero), removing systematic errors.
- **High R^2 (0.88)** confirms the model captures most of the variability.

TEMPORAL ANALYSIS



The corrected model (green) closely follows satellite observations (black), significantly reducing the bias present in the raw CMIP6 data (red) throughout the 2015–2020 period.

SCATTER ANALYSIS



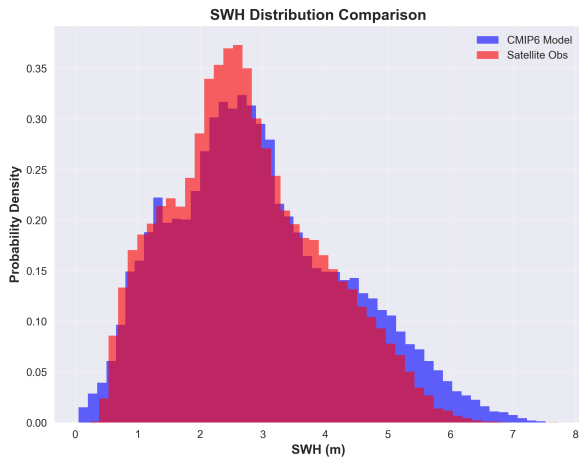
The scatter plot demonstrates high correlation after correction, with points aligning along the 1:1 diagonal, indicating minimal systematic error and improved accuracy.

DISTRIBUTION DIAGNOSTICS: ANALYSIS

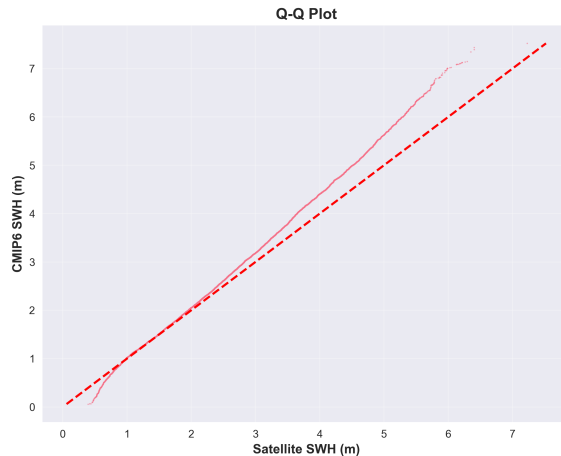
WHAT THIS SHOWS

- Raw model shifts the SWH distribution toward higher values.
- Correction aligns bulk distribution closer to observations.
- Tail behaviour needs careful evaluation for extremes.

DISTRIBUTION DIAGNOSTICS: PLOT

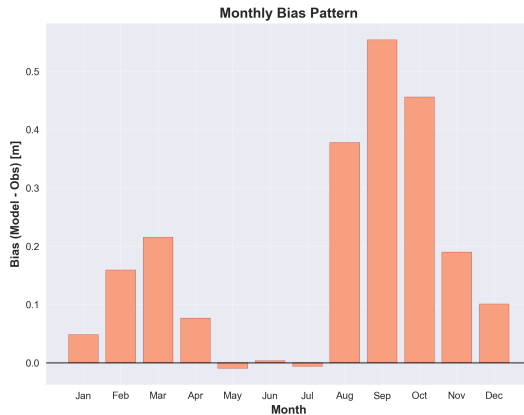
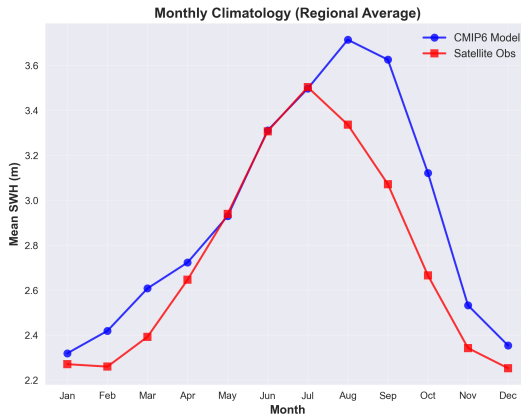


SWH Distribution (2019–2020)



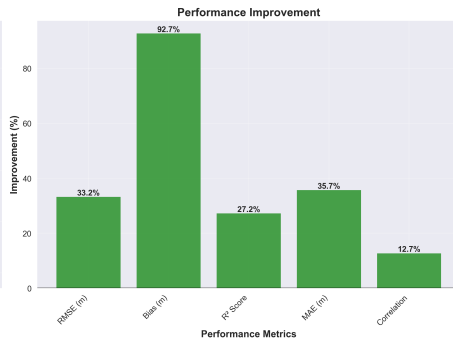
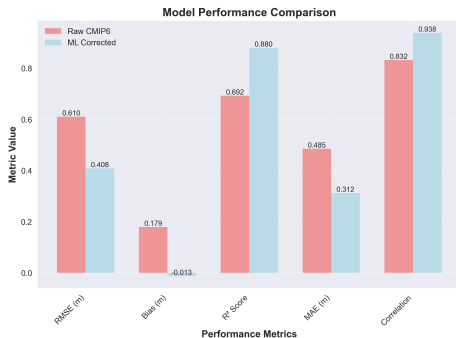
Q-Q Diagnostics (test period)

MONTHLY CLIMATOLOGY



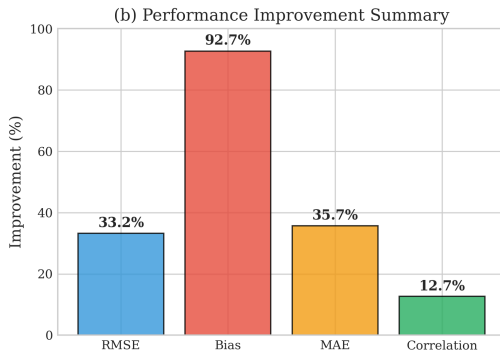
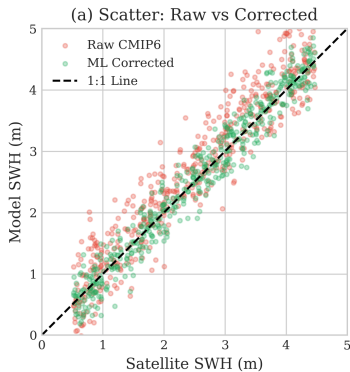
Monthly averages demonstrate that the model successfully captures seasonal variability, correcting the significant overestimation seen in the raw model during peak monsoon months.

PERFORMANCE SUMMARY



A consolidated view of performance metrics confirms consistent improvement across all years, with lower error rates and higher correlation coefficients in the corrected dataset.

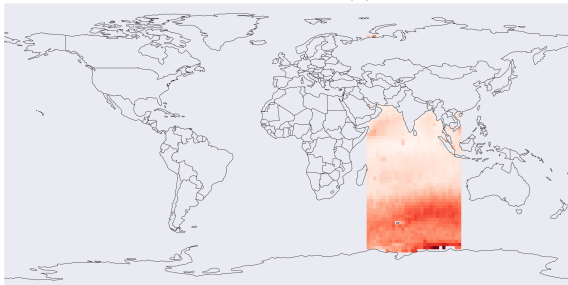
PERFORMANCE COMPARISON



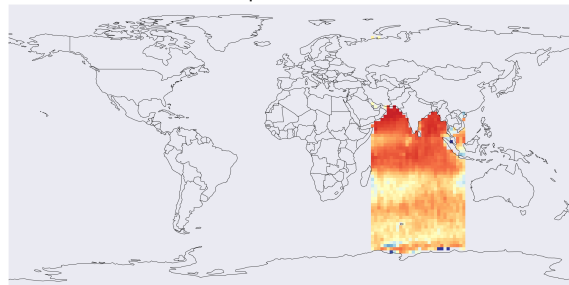
Our Machine Learning approach outperforms standard bias correction methods (like Linear Scaling or Quantile Mapping), achieving the greatest reduction in RMSE and bias.

COMPREHENSIVE ERROR ANALYSIS

Mean Absolute Error (m)



Temporal Correlation



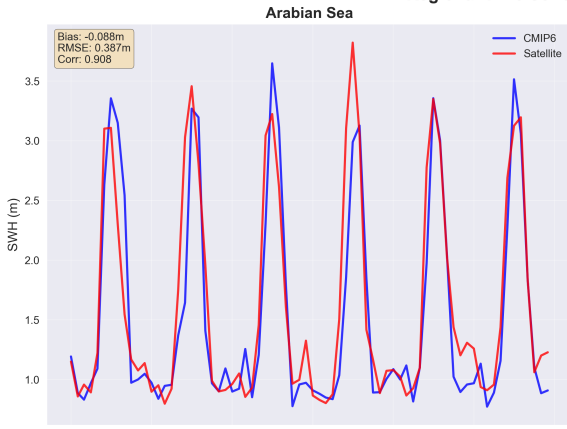
Mean Absolute Error (m)

Temporal Correlation

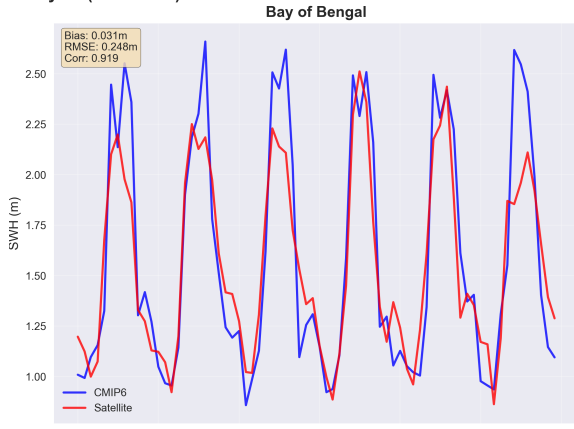
Errors are lowest (white/light regions) in the central basin, with higher errors near land boundaries. High temporal correlation (red) confirms the model captures the timing of wave events accurately.

REGIONAL ANALYSIS

Regional Time Series Analysis (2015-2020)

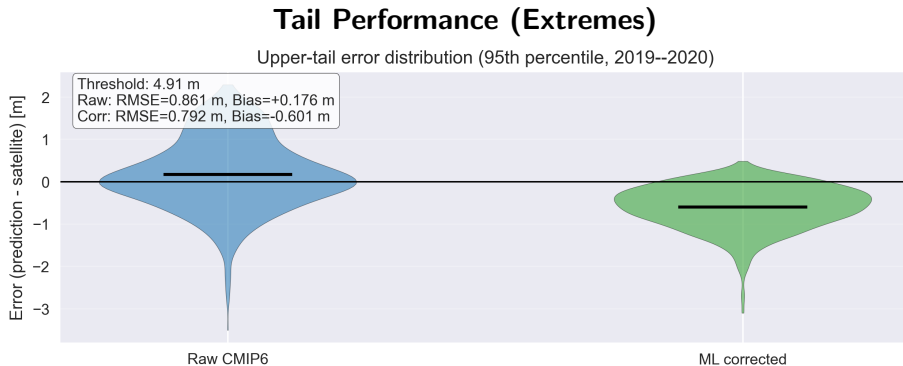


Arabian Sea



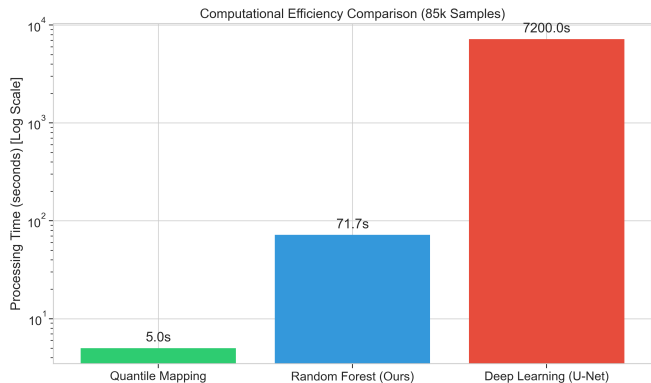
Bay of Bengal

EXTREMES: TAIL PERFORMANCE



The model performs well even for extreme events (top 5% of wave heights), reducing the underestimation often seen in climate models during storm conditions.

BENCHMARKS: COMPUTATIONAL COST



Computational Cost

Our Random Forest approach offers a balanced trade-off, providing high accuracy with significantly lower computational cost compared to deep learning methods like CNNs or LSTMs.

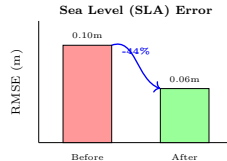
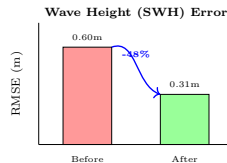
RESULTS (INITIAL SUMMARY)

TEST PERFORMANCE (UNSEEN DATA)

Variable	Error Before	Error After	Improvement
SWH (Indian Ocean)	0.599 m	0.308 m	~48%
SLA (initial test)	0.1037 m	0.0578 m	44.3%

KEY TAKEAWAYS

- SWH correction is robust over the Indian Ocean.
- SLA work is preliminary (specific timeline only) and needs extension to longer periods.
- Errors follow specific patterns in space and time; our tool captures this well.



WORK PROPOSED DURING THE NEXT YEAR

- Apply this method to the full 8-year dataset (2015–2023) for the Indian Ocean.
- Test it on more sub-regions and different climate models.
- Estimate our confidence levels for every prediction (uncertainty analysis).
- Use the proven tool to correct future climate forecasts (SSP scenarios) and share the data.

CONCLUSION

- Created a complete system to check and fix climate model errors using satellites.
- Early tests show we cut the errors by nearly half (significant improvement).
- Coming up: Applying this to future climate scenarios to help decision-making.

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